People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research

University of Abdelhamid Mehri – Constantine 2

Faculty of New Technologies of Information and Communication (NTIC)

Department of Fundamental Computing and its Applications (TLSI)

BACHELOR'S THESIS

To obtain the diploma of Bachelor degree in Computer Science

Option: Software Engineering (GL)

Adaptive Learning Platform

Realized by:

OUDJERIT Safa Maroua

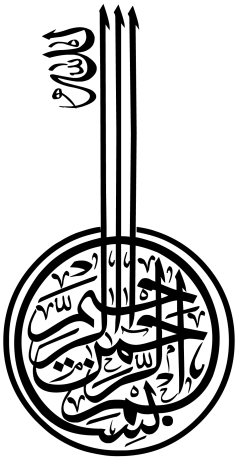
BELKHAIRI Lyna Anfel

BERRAHAL Maram

Under supervision of:

Pr.BELGUIDOUM Meriem

Mai 2026


بسم الله الرحمن الرحيم

Acknowledgments

Dedication

ملخص

تقدّم منصّات التعلّم التقليدية عادةً نفس الدروس ونفس المحتوى لجميع الطلبة دون مراعاة اختلاف مستوياتهم أو وتيرة تعلّمهم. هذا الأمر قد يجعل بعض الطلبة يشعرون بالملل، بينما قد يواجه آخرون صعوبة في الفهم لأن المحتوى لا يتكيّف مع احتياجاتهم. لذلك يركّز مشروعنا على تطوير منصة تعلّم تكيفي مدعّمة بوكيل ذكاء اصطناعي يساعد كل طالب على التعلّم حسب مستواه وتقدّمه.

تهدف المنصة إلى تقديم تجربة تعلّم مخصّصة، حيث تتغيّر الدروس والاختبارات والتوصيات بناءً على أداء الطالب وتفاعله مع النظام. يتم إنجاز هذا المشروع في إطار أكاديمي بجامعة عبد الحميد مهري ، ويعتمد على مفهوم خط إنتاج البرمجيات لجعل النظام مرنًا وسهل التوسّع بإضافة ميزات جديدة.

يسمح النظام للطلبة بإنشاء حسابات، الوصول إلى الدروس، إنجاز اختبارات بسيطة، والحصول على محتوى مناسب لمستواهم. كما يتضمّن مساعد ذكاء اصطناعي يمكنه الإجابة عن الأسئلة الشائعة، توجيه الطلبة، واقتراح دروس أو تمارين مناسبة تساعدهم على التحسّن. بصفة عامة، يهدف هذا المشروع إلى تحسين تجربة التعلّم، زيادة تفاعل الطلبة، وتوفير منصة ذكية تدعم التعلّم المخصّص لكل طالب.

الكلمات المفتاحية: التعلّم التكيفي، الذكاء الاصطناعي، منصة تعليمية، التعلّم المخصّص.

Abstract

Traditional learning platforms usually provide the same lessons and content to all students without considering their different levels or learning pace.This can make some students feel bored,while others may struggle to understand because the content does not adapt to their needs.That is why our project focuses on developing an adaptive learning platform supported by an AI Assistance(chatbot) that helps each student learn according to their level and progress.

The platform aims to offer a personalized learning experience where lessons,quizzes,and recommendations change based on the student's performance and interaction with the system.The project is carried out in an academic context at University of Abdelhamid Mehri Constantine 2 and relies on the Software Product Line (SPL) concept to make the system flexible and easy to extend with new features.

The system allows students to create accounts,access courses,complete simple quizzes, and receive content adapted to their level.It also includes an AI assistance agent that can answer frequently asked questions,guide students,and suggest suitable lessons or exercises to help them improve.

Overall,the goal of this project is to enhance the learning experience,increase student engagement,and provide a smart platform that supports personalized learning for each student.

Keywords: Adaptive learning, artificial intelligence, learning platform, personalized learning.

Résumé

Les plateformes d'apprentissage traditionnelles proposent généralement les mêmes cours et contenus à tous les étudiants, sans tenir compte de leurs niveaux ou de leur rythme d'apprentissage.Cela peut entraîner de l'ennui pour certains étudiants,tandis que d'autres peuvent rencontrer des difficultés à comprendre,car le contenu ne s'adapte pas à leurs besoins.C'est pourquoi notre projet se concentre sur le développement d'une plateforme d'apprentissage adaptative,soutenue par un agent d'intelligence artificielle,qui aide chaque étudiant à apprendre selon son niveau et sa progression.

La plateforme vise à offrir une expérience d'apprentissage personnalisée,où les cours,les tests et les recommandations évoluent en fonction des performances et des interactions de l'étudiant avec le système.Ce projet est réalisé dans un cadre académique à l'Université Abdelhamid Mehri Constantine 2 et repose sur le concept de Software Product Line (SPL) afin de rendre le système flexible et facilement extensible avec de nouvelles fonctionnalités.

Le système permet aux étudiants de créer des comptes,d'accéder aux cours,de passer des tests simples et d'obtenir un contenu adapté à leur niveau.Il inclut également un Assistant d'intelligence artificielle capable de répondre aux questions fréquentes,de guider les étudiants et de proposer des cours ou exercices appropriés pour les aider à progresser.

En général,ce projet vise à améliorer l'expérience d'apprentissage,à augmenter l'engagement des étudiants et à fournir une plateforme intelligente supportant l'apprentissage personnalisé pour chaque étudiant.

Mots clés : apprentissage adaptatif, intelligence artificielle, plateforme éducative, apprentissage personnalisé

Table of Contents

Acknowledgments ii

Dedication iii

Abstracts iv

Table of Contents vii

List of Figures ix

List of Tables x

List of Algorithms xi

General Introduction 1

I State Of The Art 5

1 State of the Art: Survey on some existing Adaptive Learning platforms 6

1.1 Overview plan about existing Adaptive Learning Systems 7

1.2 International Adaptive Learning Systems 7

1.2.1 k-12 and Higher education platforms 8

1.2.2 k-12 Adaptive Systems 8

1.2.3 Higher Education Adaptive Systems 8

1.2.4 Language Learning Platforms 9

1.2.5 Corporate and Entreprise Platforms 10

1.3 Local Adaptive Learning Systems in Algeria 10

1.3.1 FC-CAMLS 11
1.3.2 Système Multi-Agents 11
1.3.3 IAHS-TS 11
1.3.4 Adoption of International Platforms in Algeria 12
1.4 Comparative Study 12

2 Design and Development methodology **14**

3 Contribution:
Preliminary analysis and design **15**
3.1 Project Description 15
3.2 Preliminary Analysis 16
3.2.1 Requirement Analysis 16
3.2.2 Requirement specification 18

Bibliography **31**

Acronyms **34**

List of Figures

0.1 Proposed Adaptive Learning System 3

1.1 Existing Adaptive Learning Systems 7

3.1 Student Package diagram 21
3.2 Visitor Package diagram 22
3.3 Admin Package diagram 22
3.4 Teacher Package diagram 23
3.5 Jira project backlog with user stories and effort estimations 26
3.6 Jira Dashboard showing Sprint 1 29
3.7 Jira Dashboard showing Sprint 2 29
3.8 Subtask Decomposition for User Story AL-7 in Jira 30

List of Tables

1.1 Refined Comparative Study of Adaptive Learning Platforms 12

3.1 Functional needs 17
3.2 Non-Functional Needs 18
3.3 Identification of actors 19
3.4 Identification and classification of use cases 20
3.5 Product Backlog 25
3.6 Sprint Schedule: Stories, Points and Duration 27
3.7 Sprint Schedule 28

List of Algorithms

General Introduction

Project Background

Introduction

In the rapidly evolving landscape of educational technology, the need for personalized learning experiences has become increasingly critical.Traditional e-learning platforms typically employ a one-size-fits-all approach, delivering identical content to all students regardless of their individual knowledge levels, learning pace, or comprehension abilities. This standardized methodology often results in student disengagement, with some learners struggling to keep up while others become bored due to lack of challenge. The "Adaptive Learning System with AI Assistance " project, developed under the supervision of Pr Meriem Belguidoum at the University of Abdelhamid Mehri-Constantine2, addresses these pedagogical challenges by creating a dynamic learning platform that tailors educational content to each student's unique profile and performance.

Problem Statement

Current Educational Challenges

- Heterogeneous Student Populations: Classrooms consist of students with diverse backgrounds, prior knowledge, and learning speeds.
- Fixed Curriculum Delivery: Traditional systems present the same material to all learners simultaneously.
- Limited Personalization: Instructors cannot realistically provide individualized attention to every student
- Engagement Issues: Mismatched content difficulty leads to either frustration or boredom.

- Ineffective Assessment: Standard quizzes fail to identify specific knowledge gaps or adapt to student responses.

Technological Gap

While adaptive learning concepts exist in educational research, their practical implementation in academic settings remains limited.Most institutions lack:

- Customizable platforms that can evolve with curriculum changes.
- Integration of artificial intelligence for real-time assistance.
- Scalable architectures that accommodate varying feature requirements.

Objectives

- **Provide personalized learning experiences:** Tailor educational content to match each student's individual needs, pace, and knowledge level.
- **Improve student engagement:** Keep learners motivated and involved by delivering relevant and appropriately challenging content.
- **Adapt content according to quiz performance:** Automatically adjust lesson difficulty or topics based on how well students perform in assessments.
- **To integrate an AI Assistance (CHatBot):** Offer instant, intelligent assistance to answer questions and guide students throughout their learning journey.
- **Offer recommendations and adapted content to improve user performance:** Suggest personalized lessons or exercises to help students strengthen their weak areas.
- **Provide simple and friendly interface to ensure a smooth learning experience:** Design an intuitive and easy-to-navigate platform that minimizes confusion and maximizes focus on learning.

Contribution

- **1. Personalized Learning Experience:**
  The system adapts educational content to each student's individual level, pace, and performance, moving beyond the traditional one-size-fits-all model.
- **2. Integration of an AI Assistance :**
  An intelligent chatbot is embedded to assist students in real time by answering questions, offering guidance, and enhancing the overall learning process.
- **3. Adaptive Content Based on Performance:**
  Content difficulty and recommendations are automatically adjusted according to quiz results, ensuring that each learner receives appropriate challenges and support.
- **4. Software Product Line (SPL) Architecture:**
  The platform is designed as a flexible SPL, allowing core and optional features to be easily reused, extended, or customized for different educational contexts.
- **5. Improved Student Engagement:**
  By delivering relevant and adaptive content, the system helps maintain student motivation and reduces boredom or frustration caused by mismatched material.
- **6. Simple and Friendly User Interface:**
  A clean and intuitive interface ensures that students can focus on learning without technical difficulties or confusion.
- **7. Bridging the Gap Between Research and Practice:**
  The project applies theoretical concepts of adaptive learning and AI in a real-world academic setting, demonstrating their practical feasibility and benefits.

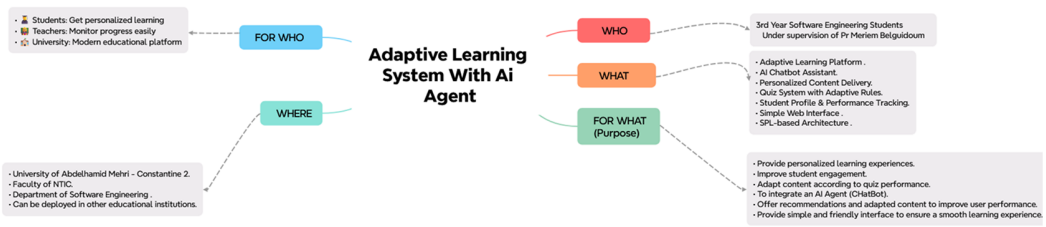


Figure 0.1: Proposed Adaptive Learning System

Document Plan

Adaptive Learning System with AI Assistance(chatbot)

This thesis is organized into two main parts, each consisting of two chapters.The structure of the document is as follows:

Part I: State of the Art

1.Chapter One: Survey on Existing Adaptive Learning Systems

This chapter provides an overview of existing adaptive learning platforms and AI-powered educational systems globally and locally. It examines platforms such as Moodle, Khan Academy, Coursera, and Duolingo, highlighting their features, adaptive mechanisms, strengths, weaknesses, and implications for the development of our proposed Adaptive Learning System with AI Assistance .

2.Chapter Two: Design and Development Methodology

This chapter discusses the methodologies and processes employed in the design and development of the Adaptive Learning System.It covers the use of Unified Modeling Language (UML) for system modeling, Software Product Line (SPL) architecture for variability management, and agile Scrum methodology for project management, detailing how these were applied throughout the project lifecycle.

Part II:Contribution

3.Chapter Three: Preliminary Analysis and Design

This chapter delves into the preliminary analysis and design phase of the Adaptive Learning System.It includes requirements analysis (functional and non-functional), specification of user stories, system architecture design, database modeling, and the design of core components including the AI Chatbot, quiz system, and content adaptation engine.

4.Chapter Four: Implementation and Validation

This final chapter focuses on the implementation of the Adaptive Learning System. It discusses the technical aspects including the choice of technologies (Node.js, MongoDB, AI APIs), coding of modules, integration of the AI Assistance , testing strategies, and validation results.The chapter details the steps taken to bring the design to fruition and evaluates the system's performance against the initial requirements.

Part I

State Of The Art

Chapter

1

State of the Art: Survey on some existing Adaptive Learning platforms

Overview

This chapter presents a structured review of the current state of adaptive learning systems worldwide and in Algeria. It explores major international platforms across K–12 education, higher education, language learning, and corporate training, analyzing their adaptive mechanisms, strengths, and limitations. It also discusses emerging Algerian research initiatives in the field and examines the adoption of international platforms within the local context. Finally, a comparative study is conducted to identify existing gaps and highlight the motivation behind the proposed system.

Introduction

In today's digital age, online learning platforms have transformed education by serving as intelligent virtual hubs for personalized learning. They play a crucial role globally and locally in modern education and professional development.

Internationally, adaptive learning systems offer flexibility across K-12, higher education, and corporate training through intelligent tutoring and real-time feedback, though they face challenges in scalability, data privacy, and effectiveness.

In Algeria, the growing interest in adaptive learning is reflected in local academic institutions developing context-specific innovative solutions, while international platforms such as Khan Academy and Duolingo continue to complement traditional education and foster self-directed learning .

These systems contribute to educational modernization and lifelong learning, but their integration requires careful consideration of infrastructure, pedagogy, and socio-cultural factors.

1.1 Overview plan about existing Adaptive Learning Systems

This survey aims to explore the landscape of existing adaptive learning systems, both globally and locally, in ordre to provide a comrehensive understanding of the current state of the art . By examining the features, services, and user experiences offered by these platforms, we seek to identify best practices and opportunities for innovation that will contribute to the advancement of adaptive learning technologies in Algeria and beyond.

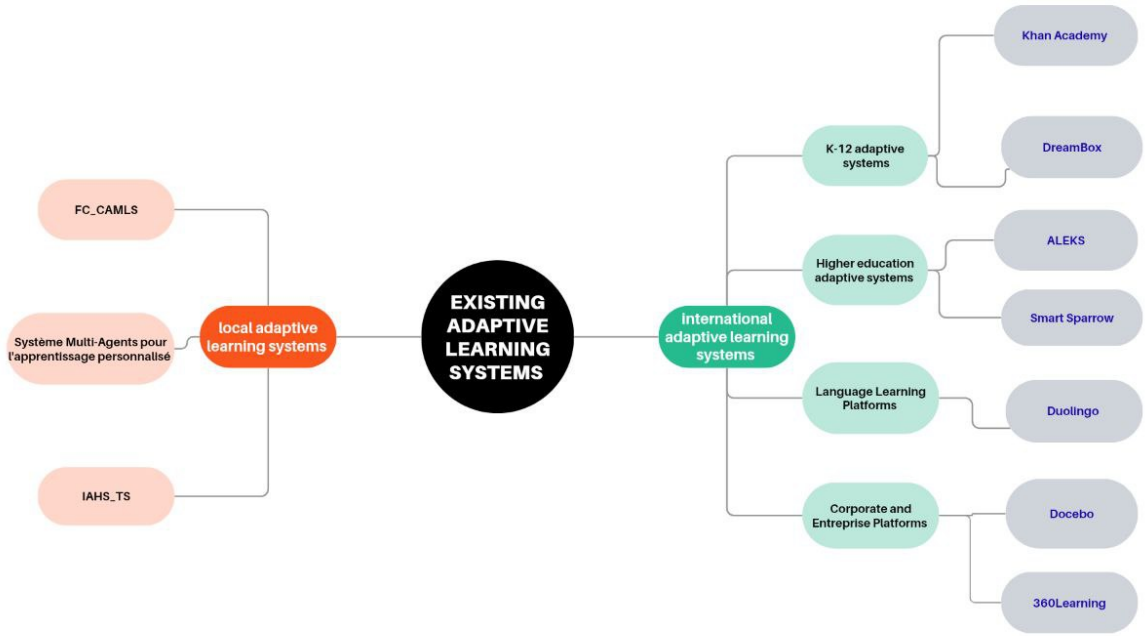


Figure 1.1: Existing Adaptive Learning Systems

1.2 International Adaptive Learning Systems

In our research, we've explored a diverse array of international adaptive learning platforms, each catering to different aspects of the educational experience. These platforms encompass a wide range of services, including K-12 education, higher education, language learning, corporate training, and specialized professional development. In this section, we'll delve into each type of platform, examining their features, services, and contributions to the EdTech industry.

1.2.1 k-12 and Higher education platforms

These platforms are designed to supplement or replace traditional curricula, focusing on deep mastery of subjects like mathematics, science, and literacy.This element explores a range of k-12 and higher education systems,highlighting key platforms,their adaptive features,and how they support personalized learning.

1.2.2 k-12 Adaptive Systems

K–12 adaptive platforms personalize learning for younger students, adjusting lessons in real-time and providing teachers with insights to guide progress. This subsection explores Khan Academy and DreamBox Learning, highlighting their adaptive features and student-centered approaches.

1.2.2.1 Khan Academy

Khan Academy [1] is a non-profit platform providing free video lessons and adaptive practice exercises, mainly for K–12 subjects such as mathematics and science. Its "Adaptive Practice" engine personalizes question difficulty to ensure mastery before progression [2]. Strengths include free access, strong STEM coverage, teacher tracking tools, and the AI tutor Khanmigo. However, it offers limited humanities content, relies heavily on video-based instruction, and requires payment for advanced AI features.

1.2.2.2 DreamBox Learning

DreamBox Learning [3] is an adaptive math program for grades K-8 that has gained recognition for its sophisticated approach to personalization. What sets DreamBox apart is its ability to analyze not just whether answers are right or wrong, but how students solve problems—tracking mouse movements, time spent on tasks, and problem-solving strategies to adjust the lesson path in real-time. Its key qualities include deeply insightful analytics, rigorous research-based math pedagogy, and a highly engaging format for young learners. However, the platform is expensive (typically licensed to schools or districts) and is limited to elementary and middle school mathematics.

1.2.3 Higher Education Adaptive Systems

Higher education platforms create tailored learning paths using diagnostic assessments and knowledge modeling. Here, ALEKS (McGraw-Hill) and Smart Sparrow are examined for their AI-driven adaptive mechanisms and support for individualized university learning.

1.2.3.1 ALEKS(McGraw-hill)

ALEKS [4] is a web-based, AI-driven assessment and learning system for mathematics and chemistry, primarily for higher education students. It uses "Knowledge Space Theory" to map exactly what a student knows and provide only content they are ready to learn. Key

strengths include highly accurate initial assessments, evidence-based STEM methodology, and a structured approach to learning. Limitations include a somewhat "robotic" and dated interface, and limited visual engagement compared to newer platforms.

1.2.3.2 Smart Sparrow

Smart Sparrow [5] is a specialized adaptive learning authoring tool designed for creating interactive and adaptive courseware, enabling educators to build "Adaptive Pathways" and "Adaptive Feedback" that direct struggling students to remedial resources while fast-tracking those who excel. Its strengths include full instructor empowerment, rich analytics into student learning patterns, and excellence in simulation-based learning that embodies the "learning by doing" philosophy. However, its weaknesses include post-acquisition uncertainty following its purchase by Pearson, raising concerns about long-term independence, and a steep learning curve that makes course design more complex than using standard LMS features [6].

Comparative Observation While K–12 systems prioritize mastery learning and behavioral analytics, higher education platforms emphasize knowledge modeling and structured diagnostic assessment. In contrast, corporate systems align personalization with workforce productivity and performance metrics rather than academic mastery.

1.2.4 Language Learning Platforms

These platforms focus specifically on language acquisition, often using gamification and spaced repetition to build vocabulary and grammar skills in an engaging manner.This subsection examines major platforms such as Duolingo,highlighting their adaptive features,strenghts, and showing how they support personalized language learning for diverse users.

1.2.4.1 Duolingo

Duolingo [7] is a mobile-first language learning platform known for its gamified approach to language acquisition. With over 500 million users worldwide [8], it is the most downloaded education app globally. Its adaptive algorithm uses AI and spaced repetition to personalize lessons, predict forgetting patterns, and adjust difficulty in real-time. The proprietary S2A3 system enables real-time question calibration, while Duolingo Max offers AI features like "Explain my Answer" and conversational "Role-plays." Key strengths include exceptional engagement through gamification, an intuitive user experience, and a powerful AI backend. However, weaknesses include criticism for achieving only surface-level proficiency, translation-heavy exercises, and gamification sometimes overshadowing actual learning goals [9].

1.2.5 Corporate and Entreprise Platforms

These platforms are designed for workforce development, focusing on upskilling employees, compliance training, and knowledge retention in professional environments.This subsection examines key corporate systems, such as Docebo and 360Learning, highlighting their adaptive features, scalability, and impact on employee skill development.

1.2.5.1 Docebo

Docebo [10] is a Learning Management System (LMS) with AI-powered capabilities designed for enterprise clients. The platform automates content tagging, recommends personalized learning paths based on user roles, and enables content upskilling through intelligent recommendations. Its key strengths include high scalability for large enterprises and excellent AI automation that reduces manual management overhead. However, the platform can be complex and expensive to set up, and its core function remains an LMS with adaptive learning as a key feature rather than its sole purpose [11].

1.2.5.2 360Learning

360Learning [12] is a collaborative learning platform focused on "social learning" that enables internal experts to create courses quickly and efficiently. The platform's AI recommends relevant peer-generated content to close skill gaps, fostering a culture of continuous learning within organizations. Its key qualities include excellence in leveraging internal company knowledge, fast course creation tools, and promotion of collaborative learning cultures. However, content quality can vary significantly if not properly moderated, and the platform is less suited for highly standardized, regulated training that requires consistent content delivery [13].

1.3 Local Adaptive Learning Systems in Algeria

This subsection provides an overview of adaptive learning initiatives in Algeria. It highlights academic projects and research-based systems developed locally, examining their innovative approaches, objectives, and contributions to personalized learning in the Algerian educational context.

Algeria has a growing academic interest in educational technology, with several university research projects exploring adaptive learning systems to address local educational needs. In light of these developments, the emergence of such academic research projects represents a promising step towards addressing the gaps in the Algerian EdTech sector. Although large-scale commercial platforms are not yet widespread, these initiatives demonstrate pioneering efforts to develop context-appropriate solutions for Algerian learners. Additionally, global platforms like Duolingo have been adopted by Algerian students and have even become subjects of academic study within the country.

Among these initiatives, FC-CAMLS, multi-agent systems from Constantine 2 University, and IAHS-TS from Annaba University have emerged as notable research contributions. While primarily academic projects rather than commercial platforms, they demonstrate the growing interest and expertise in adaptive learning technologies within Algerian institutions. This study examines their innovations, methodologies, and overall impact on the local adaptive learning landscape.

1.3.1 FC-CAMLS

FC-CAMLS (Flipped Classroom based on Context-Aware Mobile Learning System) [14] is an exploratory research project developed at an Algerian university focusing on mobile learning innovation. The system adapts educational content based on the learner's real-time feedback and context within a flipped classroom model, representing an innovative approach to combining modern pedagogy with adaptive technology for contextually relevant learning experiences. Its key contribution lies in exploring how mobile devices can be leveraged for personalized learning in the Algerian context. However, as an academic research project, it remains in the experimental phase and is not yet available as a commercial platform for widespread public use [15].

1.3.2 Système Multi-Agents

The "Système Multi-Agents pour l'Apprentissage Personnalisé" represents a collaborative project between Université Constantine 2 Abdelhamid Mehri (LIRE Laboratoire) and the Laboratoire Hubert Curien in France [16]. The system uses intelligent agents to model the learner by identifying learning styles and dynamically adapting content to suggest personalized learning paths, demonstrating a sophisticated research-driven approach that considers individual cognitive preferences [17]. Its key strength lies in its foundation in intelligent agent technology and the integration of learning styles as a personalization factor. However, like many academic projects, it remains a research prototype and has not been developed into a widely accessible platform [18].

1.3.3 IAHS-TS

IAHS-TS (Intelligent and Adaptive Hypermedia System based on Thinking Style) [19] is an intelligent hypermedia platform developed at Université Badji Mokhtar d'Annaba. The system personalizes learning resources based on two key user characteristics: "thinking style" as defined by the Herrmann Brain Dominance Instrument (HBDI) and current knowledge level. This dual-focus approach integrates cognitive science frameworks with adaptive hypermedia technology, recognizing that how learners think is as important as what they know. However, as a research project, its scope remains limited to academic experimentation rather than widespread deployment [20].

1.3.4 Adoption of International Platforms in Algeria

Beyond locally developed systems, international adaptive learning platforms have gained significant traction among Algerian learners. Duolingo, in particular, has become a subject of academic study within Algerian universities. A 2025 study published in the Algerian Scientific Journal Platform (ASJP) examined student practices and uses of Duolingo at the Department of French, University of Mostaganem, analyzing how students utilize the platform's chatbots and adaptive systems to supplement language learning. This research provides valuable insights into the adoption patterns of adaptive learning technologies in the Algerian context [21]. Similarly, Khan Academy is widely accessible and likely used by students across the country, particularly for STEM subjects, though formal studies on its usage in Algeria remain limited.

1.4 Comparative Study

In this section, we will present a table containing a comparative analysis of various adaptive learning platforms, highlighting their market positions and core differentiators in relation to our understanding of the adaptive learning landscape.

Criteria	Khan	DreamBox	ALEKS	Duolingo	Docebo	360Learning	FC-CAMLS	Multi-Agents	IAHS-TS	Moodle	Edufin
1. Adaptive Intelligence Axis											
Quiz-Based Content Adaptation	✓	✓✓	✓✓	✓✓	✓	✓	✓✓	✓✓	✓✓	✓	✓✓
Personalized Learning Path	✓	✓✓	✓✓	✓✓	✓	✓	✓✓	✓✓	✓✓	✓	✓✓
Mastery-Based Progression	✓	✓✓	✓✓	✓✓	limited	limited	✓✓	✓✓	✓✓	✓	✓✓
2. Pedagogical & Learning Axis											
Student Progress Tracking	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓
Learning Analytics	✓	✓✓	✓✓	✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓	✓
Student Feedback to Instructor	limited	X	X	X	✓✓	✓✓	N/A	N/A	N/A	✓	✓✓
3. Instructor & System Management Axis											
Instructor Dashboard	✓	✓✓	✓✓	X	✓✓	✓✓	N/A	N/A	N/A	✓✓	✓✓
Content Modification Capability	X	X	X	X	limited	✓✓	N/A	N/A	N/A	✓✓	✓✓
Performance Monitoring Tools	✓✓	✓✓	✓✓	X	✓✓	✓✓	N/A	N/A	N/A	✓✓	✓✓
4. Technical Specialization Axis											
Programming-Focused Platform	limited	X	X	X	X	X	X	X	X	limited	✓✓
Technical Skill-Based Assessment	✓	X	X	X	X	X	X	X	X	X	✓✓
5. Local & Contextual Axis											
Local Context Alignment (Algeria)	X	X	X	X	X	X	✓✓	✓✓	✓✓	X	✓✓
Arabic Language Support	limited	X	X	✓✓	X	X	✓✓	N/A	✓✓	limited	✓✓

Table 1.1: Refined Comparative Study of Adaptive Learning Platforms

Legend: ✓ = Good, ✓✓ = Excellent, X = Not available, N/A = Not applicable.

Conclusion

This chapter presented a structured review of existing adaptive learning systems at both international and local levels to establish the current state of the art. The analysis examined representative platforms across multiple educational sectors, including K-12 education (Khan Academy and DreamBox Learning), higher education (ALEKS and Smart Sparrow), language learning (Duolingo), and corporate training (Docebo and 360Learning).

The comparative investigation revealed several important observations. First, adaptive learning systems differ substantially in their personalization strategies: some are based on mastery learning models, others rely on behavioral analytics, knowledge-state modeling, intelligent tutoring, or AI-driven recommendation engines. Second, adaptation objectives vary by context: K-12 and higher education systems focus on conceptual mastery and academic progression, while corporate platforms align personalization with productivity and performance metrics. Third, although international platforms demonstrate high scalability and technological maturity, many remain costly or institutionally restricted, raising concerns about accessibility and equity.

In the Algerian context, the review highlighted emerging academic initiatives such as FC-CAMLS, the multi-agent adaptive system at Université Constantine 2, and IAHS-TS at Université Badji Mokhtar d'Annaba. Although currently research prototypes, they reflect growing research capacity and institutional engagement in adaptive learning technologies. Moreover, widespread adoption of platforms like Duolingo among Algerian learners indicates tangible demand for adaptive educational tools.

The analysis also identified gaps in the current landscape, including limited contextual adaptation to local realities, high implementation costs, insufficient pedagogical integration, and the tension between scalability and deep personalization.

Overall, this review provides a conceptual foundation for developing a context-aware adaptive learning system tailored to local needs. By integrating robust AI techniques with pedagogical coherence and socio-cultural alignment, the proposed system aims to address limitations identified in existing platforms.

The following chapter presents the modeling and development methodologies adopted in this work, including the Unified Modeling Language (UML), the Scrum agile framework, and the Unified Process (UP), which guide the system design and implementation phases

Chapter

2

Design and Development methodology

Chapter 3

Contribution: Preliminary analysis and design

Introduction

The Preliminary Analysis and Design phase represents a fundamental step in the development of our Adaptive Learning System with an AI Assistance(chatbot). During this phase, we focus on identifying functional and non-functional requirements, defining the system architecture, and modeling the main components. This structured approach ensures the creation of a scalable, maintainable, and user-centered platform capable of delivering personalized learning, adaptive content based on student performance, and intelligent support through an integrated AI Assistance(chatbot), particularly for Software Engineering students and beginners in programming.

3.1 Project Description

This project focuses on the design and development of an Adaptive Learning System (ALS) integrated with a basic AI Assistance(chatbot). The platform aims to provide personalized learning experiences by adapting educational content according to students' performance and progress.

The system includes essential features such as user registration and authentication, course access, quiz evaluation, progress tracking, and adaptive content delivery. In addition, an AI assistant is integrated to support learners by answering questions, providing guidance, and recommending relevant learning materials.

The development process follows Agile methodology, ensuring iterative implementation and continuous improvement throughout the project lifecycle. UML is adopted for system modeling to ensure clarity, consistency, and structured design.

The objective of this project is to deliver a scalable, efficient, and user-centered learning platform that enhances programming education through personalization and intelligent assistance.

3.2 Preliminary Analysis

The Preliminary Analysis phase represents a fundamental step in the development of our Adaptive Learning System (ALS) with an AI Assistance(chatbot). This phase establishes the foundation for successful system implementation by thoroughly analyzing and defining the project requirements. It mainly consists of requirement analysis and requirement specification.

During the requirement analysis stage, we identify and examine the needs of the primary users, including Software Engineering students and beginners interested in learning programming. We analyze their expectations regarding personalized learning, course access, quizzes, performance tracking, and AI-based assistance. This step allows us to clearly define the system's objectives, constraints, and scope.

Following this, the requirement specification phase translates the identified needs into structured and formal documentation. Functional and non-functional requirements are clearly defined, and UML diagrams such as use case diagrams and system models are created to describe system interactions and behavior. These specifications serve as a reference framework for the design and implementation phases.

In this section, we follow the Unified Modeling Language (UML) guidelines as described in standard UML references. The development process is aligned with Agile principles and organized into iterative sprints, allowing incremental delivery, continuous testing, and progressive refinement of the system.

3.2.1 Requirement Analysis

3.2.1.1 Functional needs

This section outlines the functional requirements of the Adaptive Learning System. These requirements define the essential services and features the platform must provide to support personalized learning, course management, assessments, adaptive content delivery, and AI-based assistance. The table below summarizes the identified functional needs of the system.

FN ID	Description	Priority
FN-01	Visitor creates an account	Highest
FN-02	Student logs in to their account	Highest
FN-03	Student logs out of their account	Highest
FN-04	Student views and edits their profile	Highest
FN-05	Display a list of available courses for the student	Highest
FN-06	Student enrolls in a course	Highest
FN-07	Student views their enrolled courses	Highest
FN-08	Student views course content (lessons, videos, text)	Highest
FN-09	Student searches for courses.	High
FN-10	Student filters courses (by level, category, or subject).	High
FN-11	Student resumes a course from where they left off	High
FN-12	Student tracks their progress in each course	High
FN-13	Student takes quizzes.	High
FN-14	Student views quiz results (correct/incorrect answers)	High
FN-15	Student submits feedback about a course	High
FN-16	Student views personalized dashboard (progress, enrolled courses, quiz scores, recommendations)	High
FN-17	Teacher views student feedback and dashboard	High
FN-18	Teacher creates quizzes for a course	High
FN-19	Teacher manages course content (add/update/delete lessons)	High
FN-20	The system adapts subsequent course content based on quiz results	Medium
FN-21	Student chats with AI chatbot to ask questions	Medium
FN-22	AI Assistance(chatbot) answers frequently asked questions (FAQ)	Medium
FN-23	Student saves favorite courses	Medium
FN-24	Student views their favorite courses	Medium
FN-25	The system notifies the student	Medium
FN-26	Student views notifications	Medium

Table 3.1: Functional needs

3.2.1.2 Non-Functional Needs

This section presents the non-functional requirements of the Adaptive Learning System. These requirements define the essential quality attributes and technical constraints that ensure system performance, reliability, security, usability, and scalability. The table below summarizes the identified non-functional requirements. The following table summarizes

all identified non-functional needs of the system:

NFN ID	Description	Priority
NFN-01	Usability: The interface must be intuitive and easy to use, even for non-technical users.	Highest
NFN-02	Page Load Time: Pages must load in less than 5 seconds for 90% of requests.	Highest
NFN-03	AI Response Time: AI Assistance(chatbot) responses must be provided in less than 5 seconds for 95% of queries.	Highest
NFN-4	Authentication: All sensitive endpoints must require authentication.	Highest
NFN-5	Modularity (SPL): Application must follow SPL principles with common core and optional features.	Highest
NFN-6	Extensibility: Architecture must allow easy addition of new features without modifying core code.	Highest
NFN-7	Concurrent Users: The system must support at least 50 concurrent users without performance degradation.	High
NFN-8	Quiz Processing: Quiz submission and scoring must complete in less than 5 seconds.	High
NFN-9	Session Management: User sessions must timeout after 30 minutes of inactivity.	High
NFN-10	Availability: The system must be available 99% of the time during core hours.	High
NFN-11	Error Handling: The system must handle errors gracefully without crashing.	High
NFN-12	Mobile Responsiveness: Application must be fully functional on mobile devices.	High
NFN-13	Cross-Platform: Application must be deployable on standard Linux/Windows servers.	Medium
NFN-14	Core functionalities must be unit tested.	Medium

Table 3.2: Non-Functional Needs

3.2.2 Requirement specification

The Requirement Specification phase defines the system in a structured manner by identifying the main actors and describing their interactions with the Adaptive Learning System. It outlines the key use cases from the user's perspective and formalizes both functional and non-functional requirements.

UML diagrams are used to model system behavior and structure, ensuring clarity, consistency, and alignment with the project objectives.

3.2.2.1 Identification of actors

Actor	type	Role
Student	Main	Uses the platform to access courses , complete quizzes and receives personalized learning content based on performance, interact with the AI Assistance(chatbot) for help.
Admin	Secondary	Manages the platform , users and educational content , ensures the system works properly.
Visitor	Main	A non register user who can view global information about platform and can create an account.
Teacher	Main	Creates and manages courses, lessons, and quizzes, monitors student performance.

Table 3.3: Identification of actors

3.2.2.2 Identification and classification of use cases

Priority and Risk Justification

Use cases were prioritized based on their importance to the system's core functionalities. Essential features such as authentication, course access, quizzes, progress tracking, and AI interaction were assigned high priority, while secondary features were given medium or low priority.

Risk levels were assessed according to technical complexity. AI integration and adaptive mechanisms present higher risks, whereas standard functionalities like login and profile management involve lower risk.

Use Case	Priority	Risks
Manage users accounts	High	Medium
Monitor platform activity	Medium	Medium
Login	High	Low
View home page	Medium	Low
View public courses	High	Low
Create account	High	Low
Contact us	Low	Low
Logout	Medium	Low
View Lessons	High	Low
Add course to favorite list	Medium	Low
Watch videos	High	Medium
Take quiz	High	Medium
View result of quiz	High	Low
Search for courses	Medium	Low
Chat with AI assistant	High	High
Track learning progress	High	Medium
View profile	Medium	Low
Edit profile	Medium	Low
Access Course Content	High	Low
Resume course	Medium	Medium
Consult score	Medium	Low
View favorite list	Medium	Low
View notifications	Medium	Low
Submit course feedback	High	Medium
Rate course	Medium	Low
Create course	High	Medium
Create quiz	High	Medium
Manage course	Medium	Low
View student Dashboard	High	Medium
View student feedback	High	Low
Enroll in courses	High	Medium

Table 3.4: Identification and classification of use cases

3.2.2.3 Use Case Diagrams

Use Case Diagrams illustrate the main functionalities of the Adaptive Learning System from the user's perspective, showing interactions between actors (e.g., students and the AI assistant) and the system. They help clarify system behavior, define supported scenarios,

and ensure that requirements are accurately modeled before design and implementation.

1.Package Diagram of Student

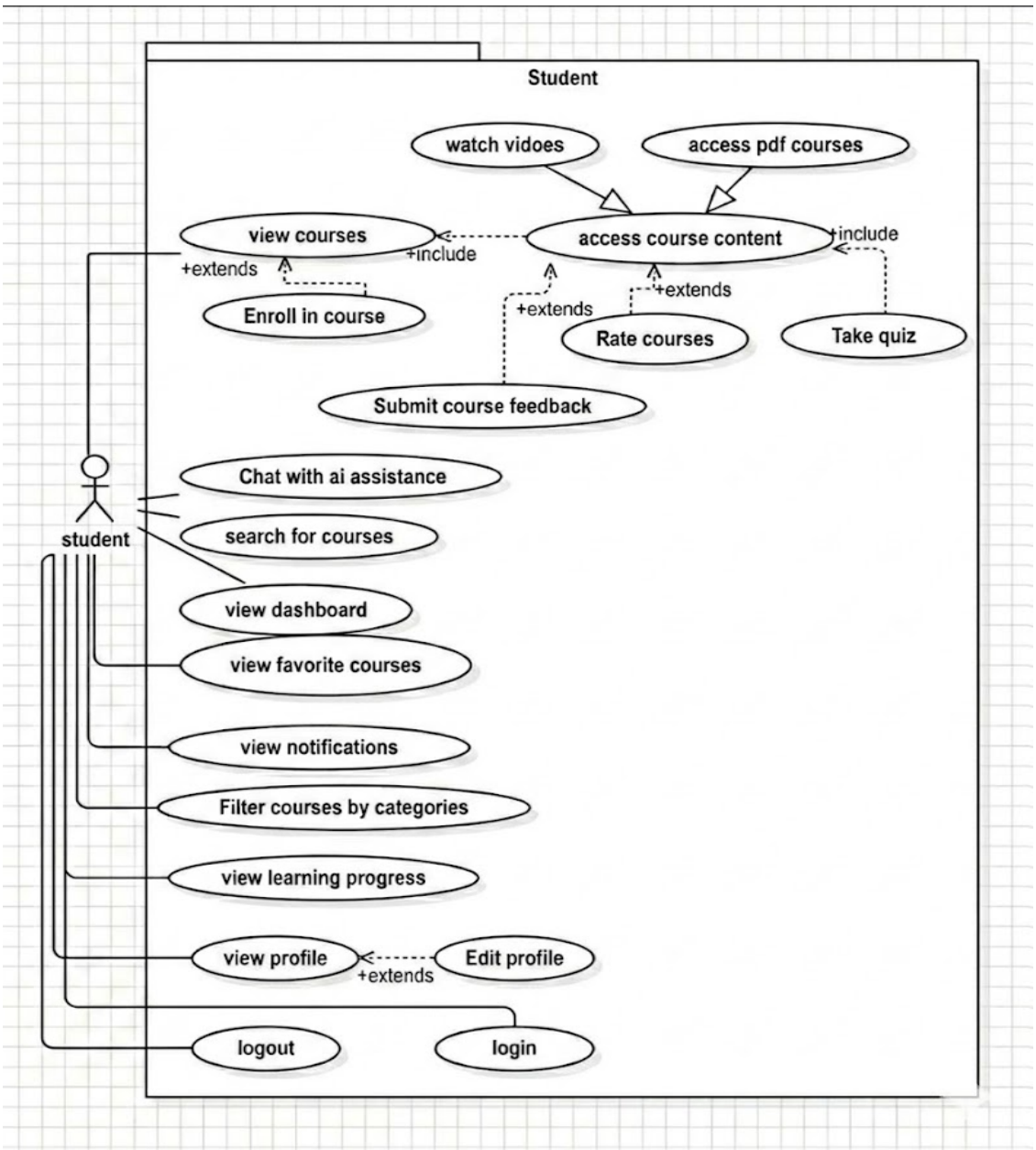


Figure 3.1: Student Package diagram

2.Package Diagram of Visitor

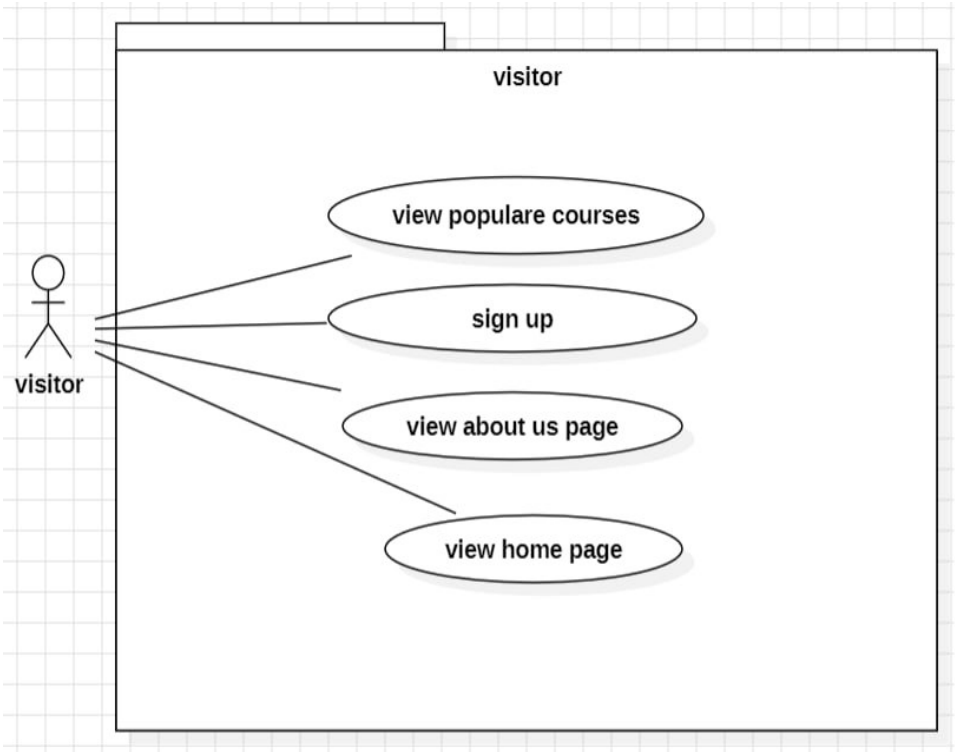


Figure 3.2: Visitor Package diagram

3.Package Diagram of Admin

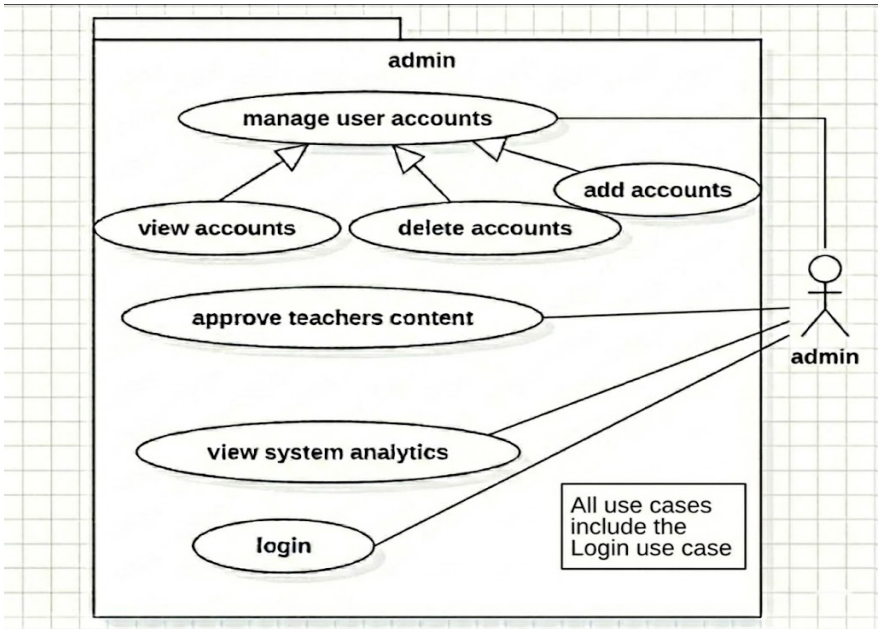


Figure 3.3: Admin Package diagram

4.Package Diagram of Teacher

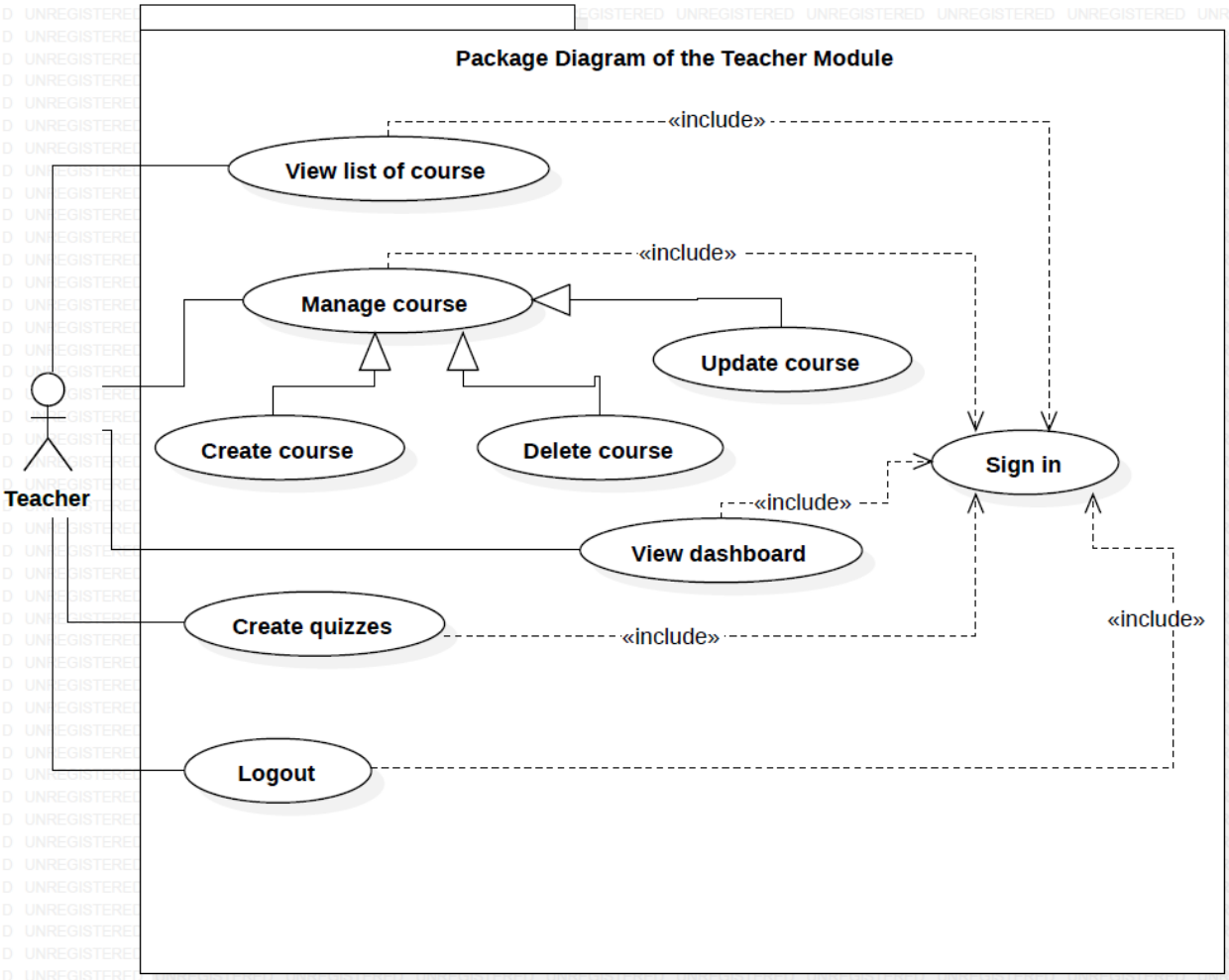


Figure 3.4: Teacher Package diagram

3.2.2.4 Project Planning

The development of the Adaptive Learning System with an AI Assistance(chatbot) was organized into iterative sprints, each addressing specific functionalities—from core features to advanced adaptive and AI components.

Agile methodology guided the process, with Jira used to manage the Product Backlog, prioritize tasks, and plan sprint activities. Each sprint included selected user stories based on priority, complexity, and project goals.

Following Unified Process principles during analysis and design ensured incremental delivery, continuous testing, and progressive system refinement.

1.Product Backlog

Story ID	User Story	Priority	Points	Status
AL-007	As a student , I want to view and edite my profile so that I can update my informations	Highest	3	Backlog
AL-008	As a visitor, I want to create an account so that access the platform.	Highest	3	Backlog
AL-009	As a student, I want to login so that I can access my account	Highest	3	Backlog
AL-010	As a teacher, I want to login so that I can access my account	Highest	3	Backlog
AL-011	As a student, I want to see a list of available courses so that I can choose one	Highest	5	Backlog
AL-012	Aa a student ,I want to enroll in a course so that I can start learning	Highest	3	Backlog
AL-013	As a student I want to view my enrolled courses so that I can continue my learning	Highest	3	Backlog
AL-014	As a student I want to access course content (pdf, videos) so I can study	Highest	5	Backlog
AL-015	As a Teacher, I want to manage courses (create, edit, delete, and publish lessons) so that I can organize and update learning content effectively	Highest	8	Backlog
AL-016	As a Teacher, I want to create and manage quizzes so that I can assess student understanding and monitor performance	Highest	8	Backlog
AL-017	As a student , I want to logout so that I can leave my account	High	2	Backlog
AL-018	As a teacher , I want to logout so that I can leave my account	High	2	Backlog
AL-019	As a student I want to search for courses so that I can quickly find a specific course	High	3	Backlog
AL-020	As a student, I want to track my progress so that I can monitor my improvement.	High	3	Backlog
AL-021	As a student I want to take quizzes at the end of each course so that I can evaluate myself	High	5	Backlog.

Story ID	User Story	Priority	Points	Status
AL-022	As a Teacher, I want to view lesson ratings so that I can evaluate content quality.	High	3	Backlog.
AL-023	As a student, I want to view quiz results so that I can understand my mistakes	High	3	Backlog
AL-024	As a student I want the system to adapt the content based on my level results so that I can receive personalized learning	High	8	Backlog
AL-025	As a Student, I want to see my dashboard (scores, progress, completion rate) so that I can track my learning journey	High	5	Backlog
AL-026	As a Teacher, I want to view the student dashboard so that I can monitor student progress and quiz performance	High	5	Backlog
AL-027	As a student I want to chat with AI assistant so that I can ask questions	Medium	5	Backlog
AL-028	As a student I want to save favorite courses so that I can access them easley	Medium	2	Backlog
AL-029	As a student I want to view my favorite courses so that I can manage them	Medium	2	Backlog
AL-030	As a student I want to receive notification so that I stay informed	Medium	3	Backlog
AL-031	As a Student, I want to rate lessons so that I can express my opinion about the content	Medium	3	Backlog
AL-032	As a Student, I want to leave feedback comments so that I can suggest improvements	Medium	3	Backlog

Table 3.5: Product Backlog

1.Implementation with Jira

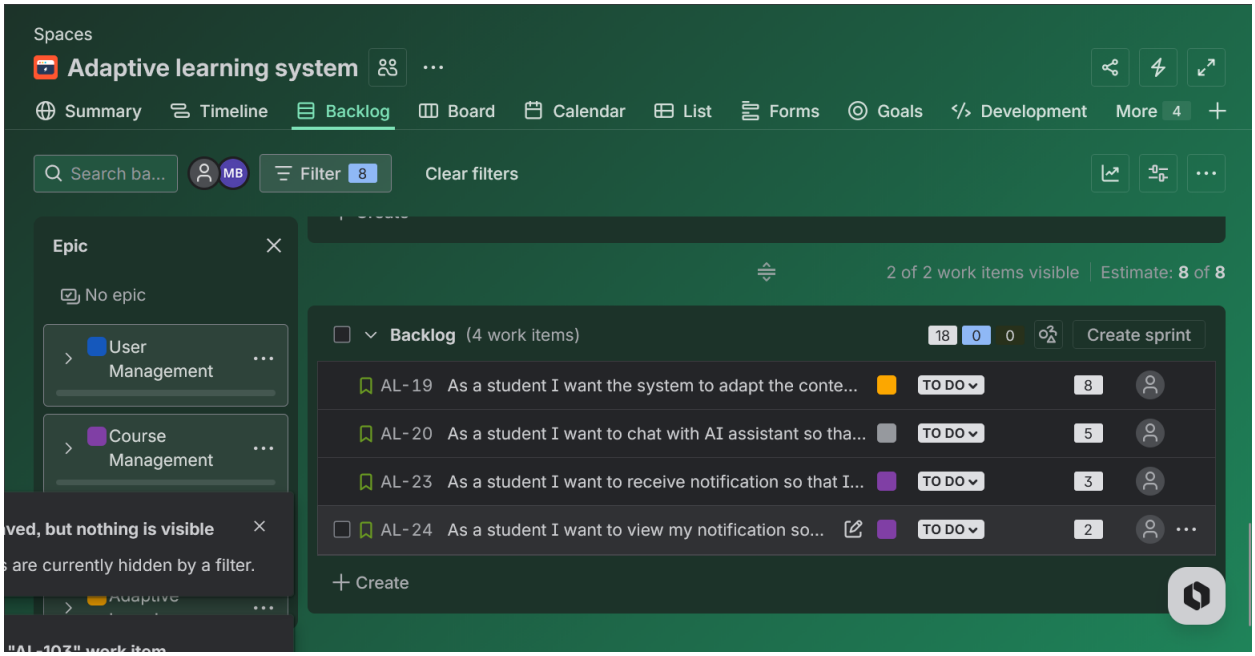


Figure 3.5: Jira project backlog with user stories and effort estimations

2. Sprint Organization & Methodology

The project follows an Agile Scrum methodology where development is organized into iterative sprints. Each sprint is planned based on the Product Backlog presented in Table above, which contains the main user stories of the Adaptive Learning System (AL-007 to AL-032).

During sprint planning, the highest priority stories are selected first, especially core functionalities such as: registration and authentication (AL-007, AL-008, AL-009), profile management (AL-007), course browsing and enrollment (AL-011, AL-012, AL-013), and course content access (AL-014). These features represent the minimal functional core of the platform.

Within each selected user story, development tasks are defined to implicitly address both functional requirements and non-functional requirements. For example, implementing the profile management feature (AL-007) involves tasks such as creating the form interface, validating inputs, and ensuring usability and accessibility of the UI. Similarly, tasks related to course access and content visualization (AL-011, AL-014) consider performance and responsiveness to guarantee smooth loading of PDFs and videos.

In later sprints, more advanced and adaptive features are developed, such as progress tracking (AL-020), quizzes and results (AL-021, AL-023), and personalized content adap-

tation based on learner performance (AL-024). These tasks directly support the adaptive learning objective of the system.

Optional and enhancement features like the AI assistant chat (AL-027), favorite courses management (AL-028, AL-029), and notifications (AL-030) are scheduled in subsequent sprints. Their implementation takes into account scalability, maintainability, and system responsiveness.

Overall, each sprint increment delivers a coherent set of functionalities while ensuring that usability, performance, security, and adaptability requirements are continuously integrated throughout the development lifecycle.

Recap of Project Setup

- **Project:** Adaptive Learning System with AI Assistance(chatbot) "Eduphin".
- **Timeline:** Feb 4, 2026 - May 7, 2026 (10 weeks, 5 sprints)
- **Team:** 3 members (1 Scrum Master, 3 Development Team members)
- **Team Members:** Maram, Lyna Anfel, Safa Maroua.
- **Tools:** Jira, Figma, GitHub, Visual Studio Code.

Sprint Schedule

Sprint	Focus	Stories	Points	Weeks	Status
Sprint 1	Core Authentication	4 stories	11	2 weeks	Planned
Sprint 2	Course Discovery	4 stories	14	2 weeks	Planned
Sprint 3	Learning Experience	4 stories	12	2 weeks	Planned
Sprint 4	Assessment	4 stories	13	2 weeks	Planned
Sprint 5	AI Features	2 stories	13	2 weeks	Planned
Total Stories: 22			**64**	**10 weeks**	

Table 3.6: Sprint Schedule: Stories, Points and Duration

Sprint Planning and User Story Distribution

Sprint	User Story	Priority	Points	Status
Sprint 1: Core Authentication & Account Management				
Sprint 1	AL-007: Create account(student and teacher)	Highest	3	Planned
Sprint 1	AL-008: login(student and teacher)	Highest	3	Planned
Sprint 1	AL-009: logout(student and teacher)	High	2	Planned
Sprint 1	AL-010: View and edit profile(student)	Highest	3	Planned
Sprint 2: Course Discovery & Enrollment				
Sprint 2	AL-011: View available courses	Highest	5	Planned
Sprint 2	AL-012: Enroll in courses	Highest	3	Planned
Sprint 2	AL-013: View enrolled courses	Highest	3	Planned
Sprint 2	AL-015: Search for courses	High	3	Planned
Sprint 3: Core Learning Experience				
Sprint 3	AL-014: Access course content	Highest	5	Planned
Sprint 3	AL-016: Track progress	High	3	Planned
Sprint 3	AL-021: Save favorite courses	Medium	2	Planned
Sprint 3	AL-022: View favorite courses	Medium	2	Planned
Sprint 4: Assessment & Feedback				
Sprint 4	AL-017: Take quizzes	High	5	Planned
Sprint 4	AL-018: View quiz results	High	3	Planned
Sprint 4	AL-023: Receive notifications	Medium	3	Planned
Sprint 4	AL-024: View notifications	Medium	2	Planned
Sprint 5: Advanced Features (AI)				
Sprint 5	AL-019: Adaptive content based on level	High	8	Planned
Sprint 5	AL-020: Chat with AI assistant	Medium	5	Planned
Total Points: 64				

Table 3.7: Sprint Schedule

Implementation with Jira

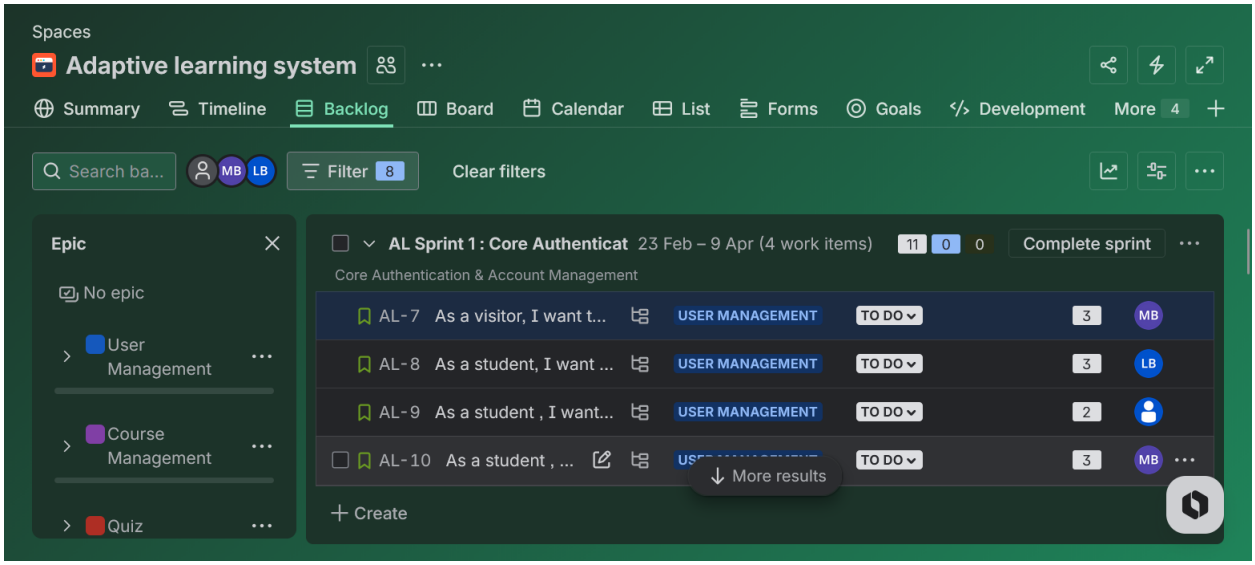


Figure 3.6: Jira Dashboard showing Sprint 1

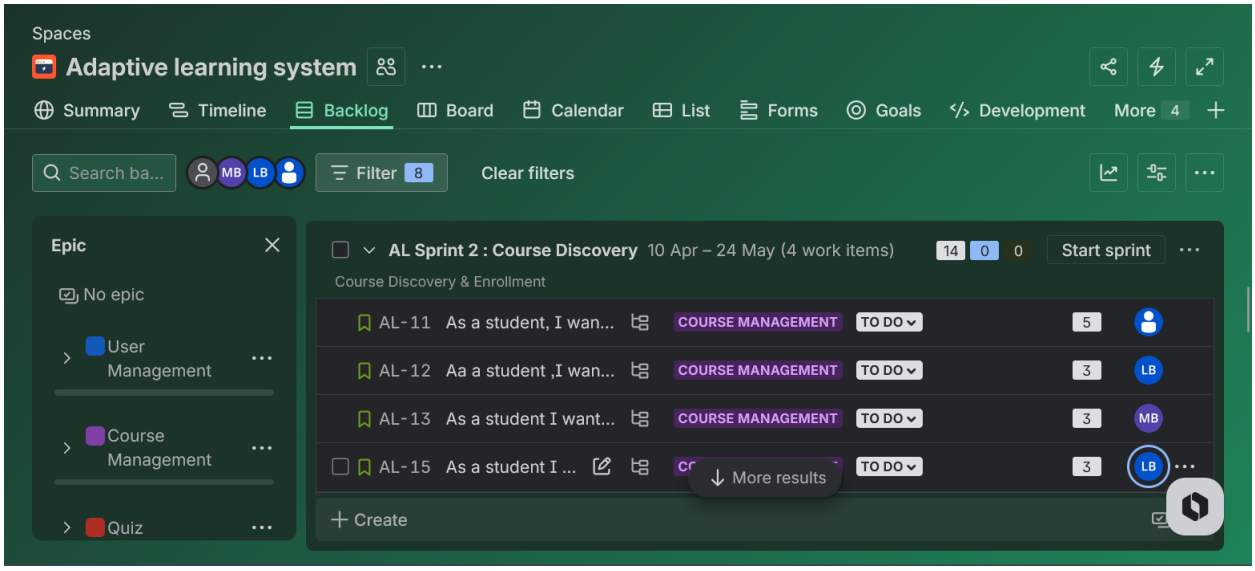


Figure 3.7: Jira Dashboard showing Sprint 2

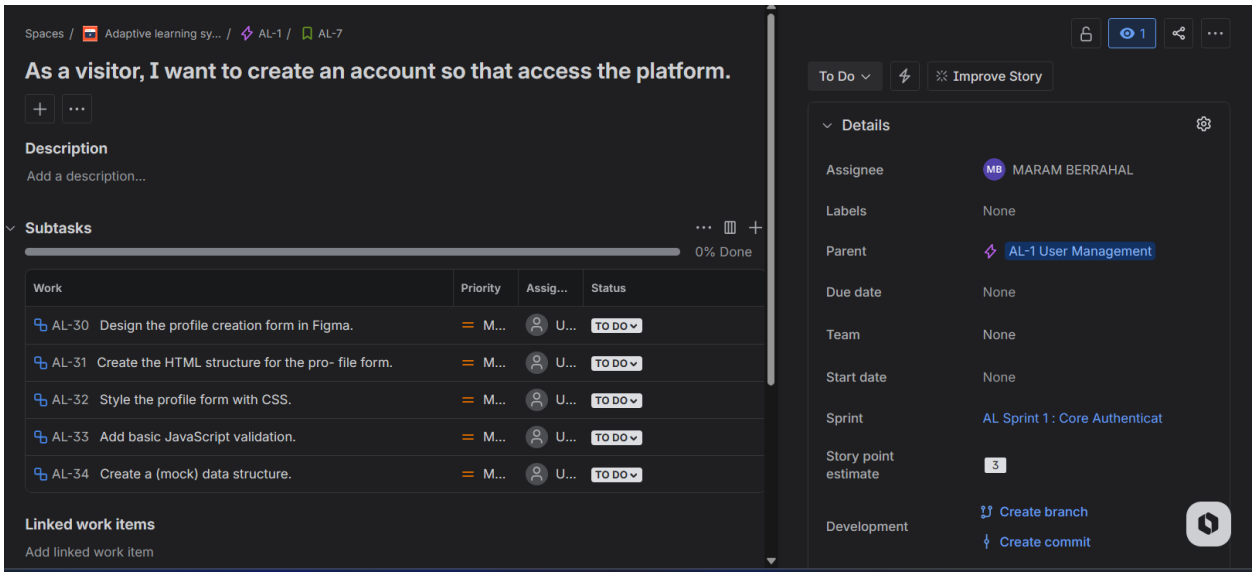


Figure 3.8: Subtask Decomposition for User Story AL-7 in Jira

Bibliography

[1] Khan Academy. Khan academy - free online courses, lessons & practice, 2026. URL https://www.khanacademy.org/. Accessed: 2026-02-22.

[2] EduGrowth. How khan academy's adaptive practice engine works, 2025. URL https://www.edugrowth.com/how-khan-academy-adaptive-practice-works. Accessed: 2026-02-22.

[3] DreamBox Learning. Official website, 2024. URL https://www.dreambox.com.

[4] McGraw Hill. Aleks - assessment and learning in knowledge spaces, 2026. URL https://www.aleks.com/. Accessed: 2026-02-22.

[5] Smart Sparrow. Our learning design platform - create courseware that engages every student, 2025. URL https://www.smartsparrow.com/platform/. Accessed: 2026-02-22.

[6] Tony Wan. Pearson bets on adaptive learning (again) with $25 macquisition of smartsparrow. EdSurge, 2020.$

[7] Duolingo. Duolingo: Learn languages for free. https://www.duolingo.com, 2026. Accessed: 2026-02-21.

[8] Chosun Media. K-content fuels global korean learning boom, surpassing chinese on duolingo. *The Chosun Daily*, April 2025. URL https://www.chosun.com/english/industry-en/2025/04/18/ZF7H2UQS7VCWZIBHTGWV3FVOKA/. Accessed: 2026-02-22.

[9] J. Sharpnack, K. Hao, P. Mulcaire, K. Bicknell, G. LaFlair, K. Yancey, and A.A. von Davier. Banditcat and autoirt: Machine learning approaches to computerized adaptive testing and item calibration. In *Large Foundation Models for Educational Assessment*, Proceedings of Machine Learning Research (PMLR), 2025. URL https://proceedings.mlr.press/v264/sharpnack25a.html.

[10] Docebo. Notre histoire | découvrez la mission de docebo, 2025. URL https://www.docebo.com/fr/company/. Accessed: 2026-02-22.

[11] Gartner Peer Insights. Docebo reviews, ratings & features 2025, February 2025. URL https://www.gartner.com/reviews/market/corporate-learning-technologies/vendor/docebo. Accessed: 2026-02-22.

[12] 360Learning. 360learning: Ai-powered collaborative learning platform, 2025. URL https://360learning.com/. Accessed: 2026-02-22.

[13] WalkMe Team. 15 best adaptive learning platforms in 2025. *WalkMe Blog*, November 2025. URL https://www.walkme.com/blog/best-adaptive-learning-platforms/. Accessed: 2026-02-22.

[14] Lamia Mahnane, Mohamed Hafidi, André Tricot, and Ouissem Benmesbah. Implementing flipped classroom that used a context aware mobile learning system into learning process. *Journal of Universal Computer Science*, 25(12):1531–1553, 2019. doi: 10.3217/jucs-025-12-1531. URL https://www.jucs.org/jucs_25_12/implementing_flipped_classroom_that.

[15] Mahnane Lamia and Hafidi Mohamed. Towards a flipped classroom based on a context-aware mobile learning system (fc-camls). In Filomena Soares, Ana Paula Lopes, Ken Brown, and Anne Uukkivi, editors, *Developing Technology Mediation in Learning Environments*, Advances in Educational Technologies and Instructional Design, page Chapter 3. IGI Global, Hershey, PA, 2020. doi: 10.4018/978-1-7998-1591-4.ch003.

[16] Maroua BENHAMLAOUI and Zakaria BENZADRI. University of constantine 2–abdelhamid mehri, lire laboratory, algeria orcid id: 0000-0002-9199-0657.

[17] Soufiane Boulehouache. *Un Système d'Apprentissage à Agents Pédagogiques Auto-Adaptatifs Basés Composants*. Doctorat en sciences, informatique, Université Abdelhamid Mehri Constantine 2, Constantine, Algérie, 2015. URL https://fr.scribd.com/document/644074812/Boulehouache-soufiane. Thèse de doctorat sur système d'apprentissage adaptatif basé agents intelligents.

[18] Mounira Bouzahzah and Ramdane Maamri. An approach for fault tolerance in multi-agent systems using learning agents. *International Journal of Intelligent Information Technologies*, 11 (3):30–44, 2015. doi: 10.4018/IJIIT.2015070103. LIRE Laboratory, University Constantine 2 - Abdelhamid Mehri, Algeria.

[19] Mahnane ML Lamia and Hafidi Mohamed. An intelligent and adaptive hypermedia system based on thinking style (iahs-ts). In *Revolutionizing Education through Web-Based Instruction*, pages 111–130. IGI Global Scientific Publishing, 2016.

[20] Mahnane Lamia and Laskri Mohamed Tayeb. Domain ontology and hermann brain dominance instrument model for personalized e-learning hypermedia system. *International Journal of Web-Based Learning and Teaching Technologies*, 7(4):1–12, 2012. doi: 10.4018/jwltt.2012100101.

[21] Aida Benhenda. Pédagogie linguistique
l're de l'ia : Chatbot interactif et systèmes adaptatifs dans duolingo. pratiques et usages étudiants en algérie. *Passerelle*, 14(1):564–579, December 2025. URL `https://asjp.cerist.dz/en/article/280145`. Université de Mostaganem, Algérie.

Acronyms

NTIC New Technologies of Information and Communication

UML Unified Modeling Language